Microplastics and nanoplastics in water treatment

Abstract:

Microplastics (MPs) and Nanoplastics (NPs) have risen since the manufacturing of plastics, which has also been aggravated by waste production of plastics, human and industrial waste. The dangers of plastic is that microorganisms form a biofilm and if the plastics are not treated then the effluent will cause major damages not just to human health but to the environment. In this literature review, we will look into many forms of treatment for the MPs and NPs although, NPs are considered difficult to be treated due to their size, density and surface properties. Current research shows that the water treatment plants can remove substantial amounts of microplastics ranging from 40% to 90%. Membrane based technologies have seen fruition by removing nanoplastics but the downside of membrane technologies is that they suffer from membrane fouling and high energy consumption. Furthermore, the research in removing the MPs and NPs have expanded but significant areas of improvements remain. Those areas include standardizing the models to detect the particles, measure and remove these particles by improving the prediction models in circumstances where MPs and NPs behave throughout the water treatment cycle. Future studies like photocatalytic degradation, magnetic nanoplastics removal and electrical chemical oxidation are still ongoing, we will also look into these future technologies to see the cost and benefits in comparison. Overall the new emerging technologies are important in upscaling the water treatment quality to address the microplastics and nanoplastics emergence.

Introduction:

Since the discovery of plastics in the late 1800's, the exposure usage of plastic has been exponential and revolutionizing industries and daily with cheap, durable material. However, this discovery has also led to significant environmental challenges and accumulation in the ecosystems. Evidence from the late 1970s shows that the dumping of fishing nets and lines led to around 105×10^3 fragments being found in the ocean (Zambello et al., n.d., #1). Furthermore, ingesting plastics potential leads to health risk as study conducted by Blackburn and Green in their study of *" The potential effects of microplastics on human health: what is known and what is unknown"* showed that the mice fed with plastics (high concentration of polyethylene) showed inflammation. In human health, the plastics enter through ingestions and inhalation, they will damage organs by inducing inflammation and immune reactions (Vethaak & Legler, 2021). In the later section, we will further study the cause of plastics damages to not only human health but also to the environment.

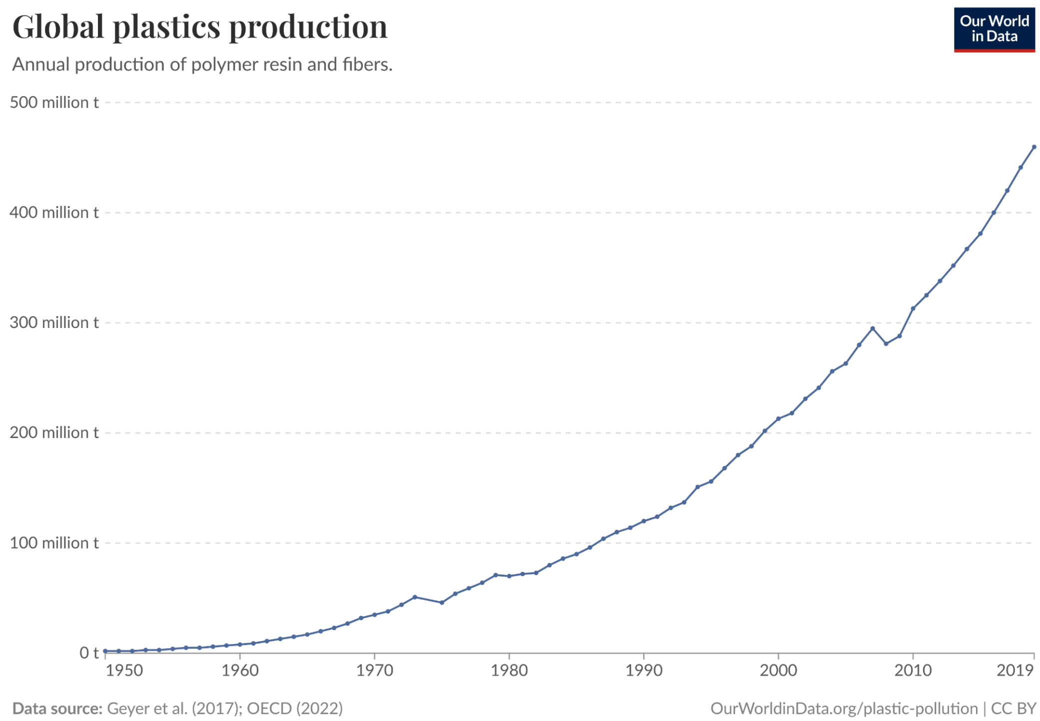


Figure 1.

Figure 1 illustrates how plastic production has grown over the years, reaching around 450 million tons globally by 2019. Since the 1950s, plastic manufacturing has gone from just a few million tons to hundreds of millions because it is cheap to make and useful for various applications. Plastics are now used everywhere in packaging, construction, and everyday items which has caused over reliance and production. This increase in global production has caused detrimental effects not just on the environment, but also in human and animal lifestyles.

As plastic production keeps increasing and plastic pollution becomes more common in both natural environments and human systems, it's becoming more important to understand how plastics affect the environment and our health. In this paper, we look at how plastics break down in nature and in wastewater systems, the risks they pose to ecosystems and people, and the different technologies used in wastewater treatment plants to remove plastic pollution. Overall, this review aims to give a clear picture of the problems caused by plastics and the efforts being made to reduce their long-term effects.

Breakdown of plastics:

The definition of plastic is that it is a material either synthetic or naturally occurring is shaped when it is soft and hardened to a given shape. Plastic are polymers meaning made up of a compound of many repeating units. The characteristics of plastics are lightweight, water and shock resistant, thermal insulating and easily mass produced. Plastic has various uses from clothing to construction usage, they are ubiquitous. Their adaptivity has allowed innovation in various industries, lowered manufacturing cost and extended product life. Most common plastics are seen from Table 1. (Chamas, 2022, #1)

Most common plastics	Density (g/cc)	Life span
Polyethylene (PE)	0.89–0.98	10-600
Polypropylene (PP)	0.83–0.92	10 - 600
Polyvinyl chloride (PVC)	1.16–1.58	50-150
Polyethylene terephthalate (PET)	0.96–1.45	450
Polystyrene (PS)	1.04–1.1	50

Table 1.

Natural breakdown of plastic:

The breakdown of plastic is a long tedious and multistage process driven by the various combination factors such as physical, environmental, and biological forces. The majority of the plastics are synthetic polymers with strong chemical bonds making them inherently resistant to degradation. Their breakdowns involve four overlapping stages, fragmentation, depolymerization, mineralisation and assimilation with the microorganisms.

1. **Fragmentation:** large pieces of plastics break down due to physical forces such as sunlight, heat, wind and mechanical processes.
2. **Depolymerization:** Long polymer chains are chemically broken down by environmental or microbial processes into shorter chains.
3. **Mineralisation:** Simpler molecules are converted into basic inorganic materials such as carbon dioxide, water or methane, depending on aerobic or anaerobic processes.
4. **Assimilation**: Micro-organisms like fungi can absorb these products through their cell membranes. Once inside, the metabolic are processed through:
    a. TCA cycle for energy production.
    b. Anabolic pathways for creating biomass production.

Breakdown of Polyvinyl Chloride (PVC):

Polyvinyl Chloride (PVC) is commonly used synthetic plastic in the pipes, packaging and medical equipment. It is chemically made up of high chlorine content about 56%, this allows the PVC to have high durability and chemical resistance properties for which they are widely used. However, these chemical properties are also the reason they are resistant to natural degradation, mainly microbial biodegradation.

PVC in comparison to other biodegradable plastics such as PET or polylactic acids lacks hydrolysable bonds, such as amides or esters that gives less access to microbial enzymes. Furthermore, microbial enzymes cannot break down PVC due to additional elements such as stabilizers and flame retardants that pose further challenges. However, only certain microorganisms have shown limited prowess in degrading the PVC such as *Pseudomonas citronellolis* and *Bacillus subtilis* have shown surface level biodegradability of PVC.

The mechanism of breakdown of PVC is initiated by dehydrocholorination process by breakdown of hydrogen chloride (HCI) is removed from the polymer chain. This process can be enhanced by physical environment processes such as UV light exposure or heat that would weaken the polymer chain. After the breakdown, oxidate enzymes can be introduced to the process like pre-oxidases to facilitate further breakdown. Nonetheless, this process is extremely slow and inefficient and full mineralization of PVC has not been seen.

Overall, SEM analysis showed that introduction of biofilm-forming bacteria from the marine environment showcased results such as grooves and cracks on the treated surface of PVC.

Breakdown of Polyethylene (PE):

Same as the PVC, Polyethylene (PE) also does not contain any amides or ester making the degradation from microbial standpoint really challenging. It is also mostly applied plastics such as in packaging films, containers and plastic bags. PE is made of a long chain of polymers -C-C chain making it highly durable and resistant to degradation. Studies have shown that specific microbial strains are able to interact with the PE especially after chemical and environmental pre-treatment. For example, UV radiation, heat or oxidative chemical exposure to PE surfaces introduce oxygen containing groups such as carbonyls or hydroxyls in the group allowing the surface to be exposed to higher surface energy promoting microbial adherence. But with the grain of salt, it does not fully degrade the polymer completely but initiates oxidative degradation, that helps increase the microbial action at the latter stages. (Carr et al., 2016)

A range of microbial species, *Pseudomonas, Bacillus, and Rhodococcus* as well as *fungi like Aspergillus and Penicillium,* have shown to play in PE degradation under specific conditions. Microbials form biofilm on the plastic surface and result in weight loss, surface pitting, or reduction in molecular weight. Furthermore, these are limited degradation and complete degradation of PE is rare.

Breakdown of Polyethylene terephthalate (PET):

Polyethylene tetraphthalate (PET) is a semi-aromatic polymer made from teraphthalic acid (TA) and ethylene glycol (EG), that are used in the making of beverage bottles, textiles, and food packaging. PET has ester linkages theoretically make PET more degradable compared to other polymers. However, PET is highly resistant due to high crystallinity and molecular weight.

One of the most important breakthroughs about PET came through a bacteria called *Ideonella sakaiensis*, that has two key important enzymes called PETase and MHETase. PETase breaks down hydrocarbons into Mono-(2-hydroxyethyl) terephthalate (MHET) and MHETase converts MHET into TPA and EG that assimilates with the microbial metabolic pathways. Further research has shown that to enhance PET degradation it is possible by three factors as follows:

- Thermal treatment to reduce the crystallization structure for later breakdown
- Alkaline hydrolysis to increase surface adherence for microbials
- UV radiation to oxidise the surface of the polymer

In depth analysis has shown that integrating consortia of enzymes may pave the path for scalable biodegradable PET. However, the degradation process of PET still remains slow but it leads us to opportunities such as closed loop PET recycling to convert waste monomers to polymerization, and for that reason the opportunities are endless once PET can be degraded naturally. (Chamas, 2022)

Importance of natural breakdown

The plastics enter the WWTP facilities as microplastics or nanoplastics often from household waste, and storm run-offs. Traditional WWTPS are not well equipped to completely remove the said micro and nanoplastics, allowing plastics to travel through filtration systems and into the marine environments. Additionally, it is also possible during sludge settlement that plastics will reduce the sludge quality and lead to potential release to the soil. Furthermore, the plastics act as embodiment of pollutants and pathogens of leading risk to environment and human health. Therefore it is highly important for microplastics to go through an economical natural breakdown or pre-treatment process to reduce the load of persistence in effluent and sludge. Developing future research in bio-technology in breakdown of plastics can align more towards sustainable wastewater management and improve efficiency. (Carr et al., 2016)

Microplastics removal in DWTP and WWTPs:

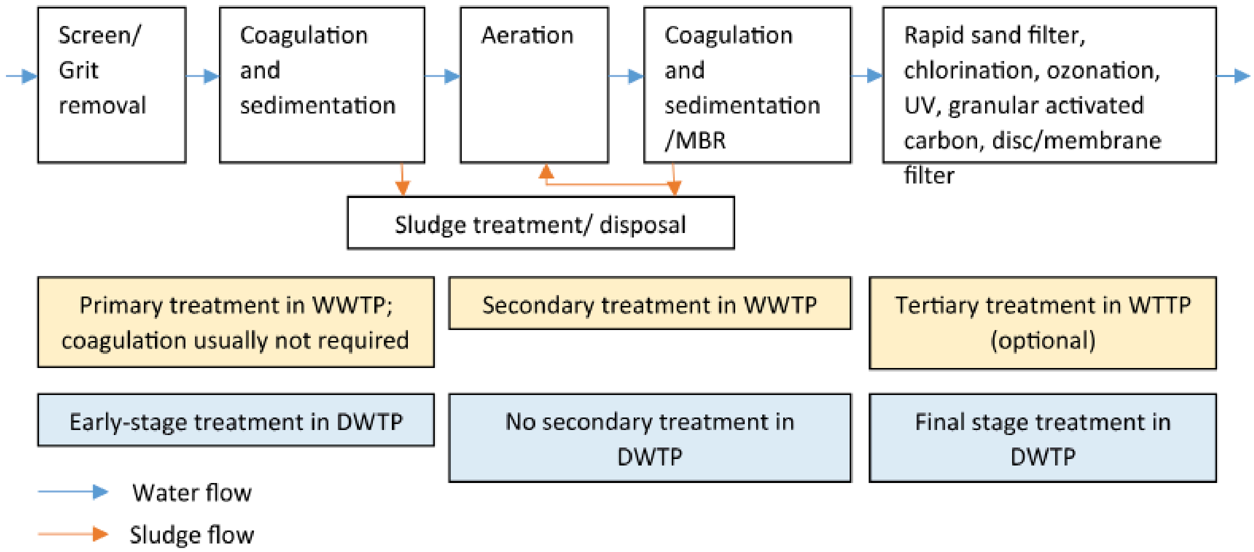


Figure 2. Stages of drinking water and wastewater treatment plant (Carr et al., 2016)

Conventional Wastewater treatment plant:

Conventional WWTP usually consists of three stages: primary, secondary and sometimes tertiary treatment that are designed to remove contaminants and solids from wastewater. The primary treatment begins with large debris collected by grit chambers that settle heavy inorganic particles like sand and gravel. After the grit removal, the wastewater flows into primary clarifiers, this is where suspended solids settle by gravity. At this stage, coagulants and flocculants are added to enhance the sedimentation process and form what is a primary sludge.

In the secondary treatment, the removal of dissolved organic matter using biological processes such as the active sludge process is the most common method used in WWTP. The aeration tank where air is pumped to support microbial activity leads microbial to degrade organic pollutants, reducing the water's biochemical oxygen demand (BOD). The product of this mixture is then transferred to a secondary clarifier, where microbial biomass settles. Portion of the sludge is sent back to aeration tank, and the rest is disposal, variation of the process includes membrane bioreactors (MBRS) and the anaerobic anoxic aerobic (A2/O) systems. MBRS replaces the secondary clarifiers with ultrafiltration members for improved separation, while A2/O systems enhance nutrient removal through anaerobic and aerobic processes.

The tertiary treatment focuses on removing residual suspended solids and dissolved substances. Those techniques include sand filtration, disc filters, biologically active filters, and rarely, granular activated carbon (GAC). Additionally, membrane technologies, such as ultrafiltration and reverse osmosis can be used for better effluent quality in advanced facilities.

Disinfection methods such as ozonation, chlorination, or UV treatment can be used before the final discharge of the effluent.

While primary and secondary have shown as per the paper by (Carr et al., 2016) remove up to 99.9% of the microplastics however, it becomes a different story when biofilm formation and sampling strategies are different as per (Shima et al., 2017) affecting the removal efficiency. Studies as per Carr et al have produced results in which microplastics were found to be at 0.0007 microplastics per litre in secondary effluent but other recorded results were about 0.47 particles/L adding the idea that various technology and methods affect the microplastics removal.

In the study conducted in "*Solutions to microplastic pollution – removal of microplastics from wastewater effluent with advanced wastewater treatment technologies."* by (Talvite & Mikola, 2017) compared the effectiveness of removal of microplastic in their final tertiary process. This tertiary process includes membrane bioreactors (MBR), discfilters, dissolved air filtration (DAF) and rapid sand filtration. The result showed that MBR that uses submerged membrane and ultrafiltration achieved 99.9% efficiency in removing the microplastics ranging in size from 20 µm to 300 µm. Discfilters on the other hand, trapping particles with filter panels showed results in removing microplastics from 30% to 98.9% depending on the condition of sludge cake formation. DAF introduces air bubbles allowing the solids to be suspended upwards as well as microplastics showcased the efficiency of removal of microplastics by 95% rate. Rapid sand filters contain layered graves and quartz removed about 95% of targeted suspended solids and nutrients. Furthermore, these results were also reassessed in "*Identification of microplastic in effluents of waste water treatment plants using focal plane array-based micro-fourier-transform infrared imaging*" by (Mintenig & Loder, 2017) also reported that the tertiary treatment composed of tertiary pile fabrication removed around 93% of microplastics ranging in size of <500 µm. Lastly, the study conducted in "*A study on characteristics of microplastic in wastewater of South Korea: identification, quantification, and fate of microplastics during treatment process*" by (Hidayaturrahman & Gwan Lee, 2019) compared tertiary treatment of microplastics through coagulation, ozone, membrane disc filter and rapid sand filter in South Korea WWTP. The results were in, coagulation removed between 92.2% and 95.7% microplastics. Ozone, membrane discfilters and rapid sand filtration removed about 99.1%, 99.2% and 98.6% as compared to Talvite research study.

Overall, the main premise of the conventional WWTP is that during the tertiary treatment points out high removal rate of removing microplastics in WWTP especially membrane and oxidation based treatments. The consistent high rates removal observed in MBR and oxidation treatment highlights the integration of such technologies in removing microplastics.

Conventional Drinking water treatment plant:

Although the drinking water plant has multiple stages such as coagulation, flocculation, sedimentation and sand filtration, it lacks the secondary process that is removal of organic material. Majority of the studies have shown the removal of microplastics are removed through primarily physical separation being activated carbon, sand filtration capture, and aggregation into the flocs. In the full scale observation of DWTP, the removal of microplastics is seen to be about 70% to 80% remaining particles left behind are < 10µm inferring that the smaller particulates are harder to capture and filtrate. Furthermore, testing with $AlCl_3 \cdot 6H_2O$, have shown effectiveness of removing PE from the water as compared to $FeCl_3 \bullet 6H_2O$ and what is surprising is that the efficiency increased as PE diameter decreased from 5000 to 50µm as assessed in the study by (Ma & Ding, 2019). Even though increasing the dosage of coagulant $AlCl_3 \cdot 6H_2O$ only elate little bit of efficiency in removing greater diameter of PE; however, addition of anionic polyacrylamide with $FeCl_3 \bullet 6H_2O$ showcased removal of PE < 500 µm about 90.9%.

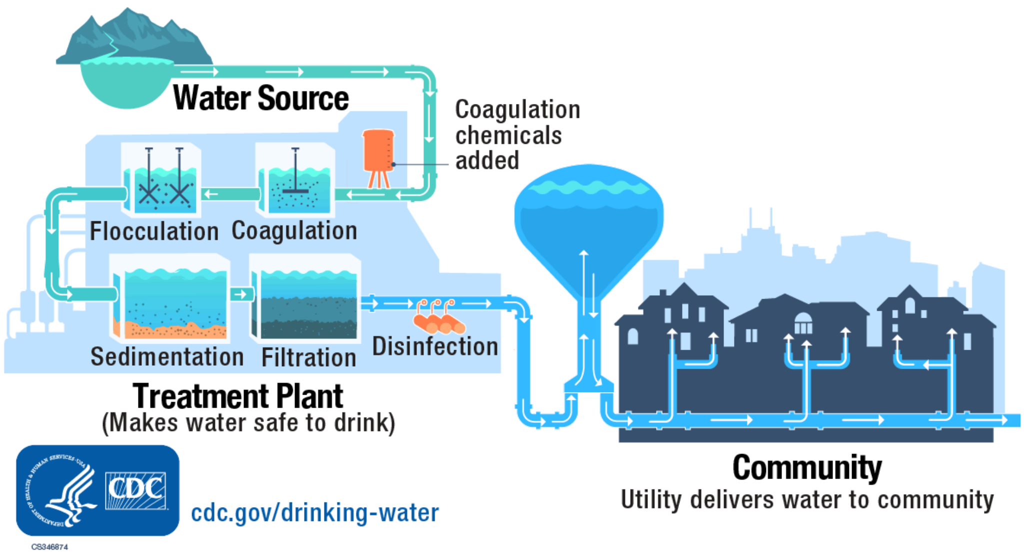


Figure 3. CDC highlight of drinking water

Further testing with the conventional coagulation serum $KAl(SO_4)_2 \bullet 12H_2O$ injected to water treatment showed capability of removing the oxide colloid, and direct removal of the plastic sphere with diameter of 1-5µm. Increasing the dosage did not have any effect on removing the PE microfibers at 20mg/L even though it was well known to remove the PE microfibers. In contrast, the $AlCl_3 \cdot 6H_2O$ removed diameter of microplastic size of 45-53µm to efficiency of 13.6%. In addition, sand filtration removed about 29% to 44% of microplastic of sizes > 5µm and without the advanced treatment, the overall efficiency of removing microplastics was about 58.9% to 75% according to research by (Pivokonsky et al., 2018). Finally, the evidence points out the DWTP conventional and advanced treatment removes the microplastics through separation and not breakdown and limited oxidation is insufficient in breaking down the microplastic.

Membrane waste water treatment plant:

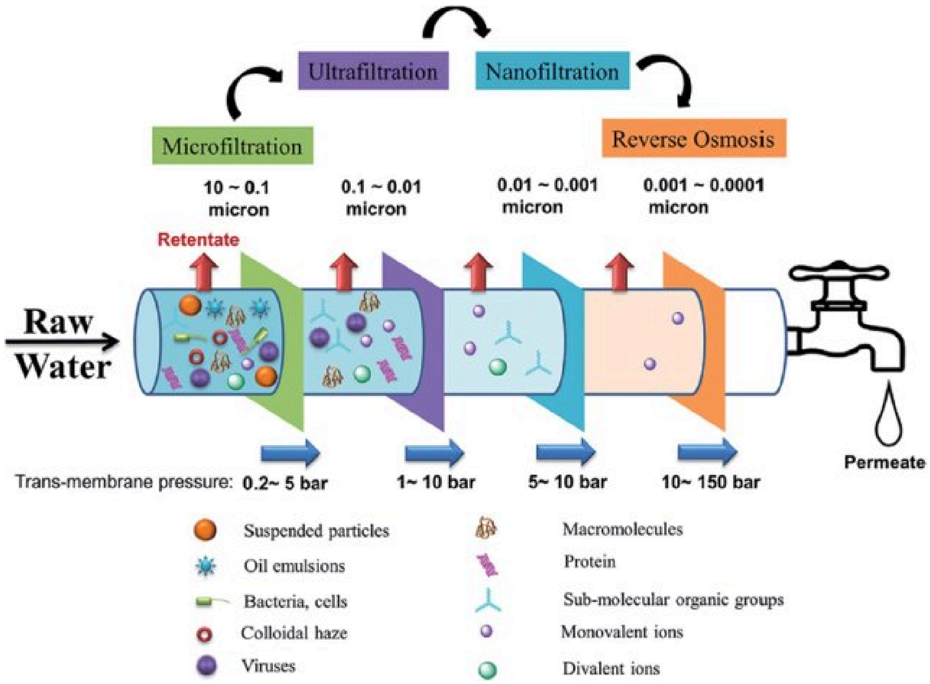


Figure 4. Membrane water treatment plant (Shima et al., 2017)

The membrane water treatment is also used in the tertiary treatment of secondary effluents in the wastewater treatment plants, for example, the use of reverse osmosis has been used in Australia for the removal of the microplastics from 0.28 per liter to 0.21 per liter (Shima et al., 2017). Although, it is astonishing to see this result despite the smaller pore size of the RO membrane than that ultrafiltrate membrane. One of the assumptions is that irregularities with the pore size of the membrane and minor defects might have led to this type of the result. There was also comparison between the use of advanced MBR and activated sludge process, the results showed that the use of MBR had higher removal of plastics efficiency by (0.4 ± 0.1 microplastics/L effluent) than the traditional usage of activated sludge (0.1± 0.4 microplastics/L effluent). From the various studies conducted by different authors such as Lares, Lv et al, Simon, and Michielssen, there is general consensus that the use of advanced MBR has resulted in common 99.4% use of micro plastics removal with removal efficiency between (> 5000µm to < 250µm). There were methods deployed in the research *"Efficiency and effectiveness of a low-cost self cleaning microplastic filtering system for wastewater treatment plants"* by (Beljanski, 2016), the test had a low cost gravity powered filtration system that can recommend optical filter type, angle and pressure. In addition, two filters were compared in the studies, 2D filter with 80µm screen and 3D with the same screen size but thicker multiscreen. The results showcased that the 3D filter was able to retain the majority of the microplastics as compared to the 2D filter. Although the 2D filter showed quicker clogging, it did better on removal of microplastics during backwashing. Backwashing to be done in 1.6kPa and 3.68kPa could remove up to 98% of microplastics but as the process was done in low pressure systems, backwashing will require longer duration.

Membrane in drinking water treatment plant:

The membrane filters used in ultrafiltration with membrane pore size of 30 nm were investigated after the sedimentation process for drinking of water. Membrane fouling was seen due to cake formation and aggregated by floc and PE particles. Microplastics that got removed in the membrane reduced the removal of organic matter and ammonia by the MBR from 50% to 40% respectively. In addition, the cake formation was aggravated by the smaller packed PE and flocs that got developed overtime with 0.5 mM $AlCl_3 \cdot 6H_2O$ added. The results also showed that the microplastics that have concentration of 10particles/L affected the performance of the MBR that usually have the efficiency between 80-90% of removal of organic matter and ammonia. (Beljanski, 2016)

It is important to see coagulant dosage of $AlCl_3 \cdot 6H_2O$ can lead to formation of smaller, denser packed flocs that may increase or lower the cake layer. Additionally, the periodic membrane cleaning and fouling control strategies like surface grafting, or chemical cleaning agents that will limit the accumulation of the microplastics on the membrane in terms increase the efficiency of removing the organic matter. (Beljanski, 2016)

Literature review of global wastewater treatment for microplastic:

In the research *"A review of the removal of microplastics in global wastewater treatment plants: Characteristics and mechanisms"* by Weiy Liu and his peers, examined the removal mechanism of microplastics in global water treatment plants by scouring the internet and other databases that resulted close examination of 38 WWTPs across 11 countries.

The methodology looked into 23 papers that were assessed based on abstracts, tables and figures for meta-analysis. Another criteria for the methodology of the paper was to see to include influent and effluent microplastics abundances, treatment process and microplastics characteristics. The microplastic size was standardised for the meta-analysis 100µm, 500µm, 1mm and 5mm. A quantitative meta-analysis was conducted using R-software and microplastic removal data from each independent study were analyzed to compare different technologies. Risk ratio was selected as the main variable to quantify microplastic removal efficiency in which RR less than 1 indicated effective microplastic removal, with lower RR values representing higher removal efficiency. $RR = At/Ae$ where At and Ae denote experimental and control group probabilities. In addition the study conducted the analysis on primary, secondary and tertiary treatment processes of microplastics removal and then compared the efficiency between all these processes. (Liu et al., n.d., #10)

The characteristics of the microplastics in the study displayed about nine different microplastics shapes in WWTP both in influent and effluent, where fibres reported to be 91.32% consisting of the major shapes in WWTPs. The detection of different plastics shapes from the study were as follows:

Shape	Detection Times
Fiber	12
Fragment	11
Film	9
Pellet	7
Foam	4
Particle	3
Ellipse	1
Line	1
Flake	1

Table 2. Shape and detection

Furthermore, the distribution of microplastics size in WWTPs were smaller than 1mm that ranged from 65.0% to 86.9 % in the influent and effluent was 81% to 91%. After all the rigorous process of removing plastics, the subsequent plastics were ingested by either plankton, filter-feeding organisms, or fishes that can cause toxicological effects in the organism. The final characteristic is the polymer type, as discussed at the start of this paper, there are various kinds of polymers and with their chemical attributes making it harder to break down and assimilate. In the research study conducted, it was found that 29 polymers were detected where PE was about 64% and the lowest one was EP around 3.4%.

The grouping of the primary, secondary, and tertiary process of WWTPS were considered as follows:

1. **Primary**: Primary, Grit and grease, combination of both grit and primary setting.
2. **Secondary**: A2O, Biofilter and Bioreactor
3. **Tertiary**: Advanced oxidation and Filter

The average rated ratio (RR) was calculated as follows:

1. Primary treatment: 0.40
2. Secondary treatment: 0.39
3. Tertiary: 0.48

Removal efficiencies	Risk Ratio
Biofiler	0.32
Filters such as UF, RS and DF	0.33
Primary settling	0.39
Bioreactors	0.41
Grit and grease removal with primary settling	0.42
Advanced oxidation process	0.56
A2O process	0.73
Grit and grease removal alone	0.61

Table 3.

As seen the filter based technologies achieved the highest removal efficiencies and the lowest was found in the tertiary side being A2O process. The reason for A2O to be less effective might be because of the re-introduction of the sludge which brings back microplastics. In addition, the influence of particle size also matters the most as per the study it follows:

1. Small microplastics(<0.5mm): They were removed during secondary treatment,
    a. Primary RR: 0.70
    b. Secondary RR: 0.48
2. Medium size microplastics (0.5 - 1mm) were better removed during primary treatment:
    a. Primary RR: 0.31
    b. Secondary RR: 0.74
3. Large microplastics(1-5mm) showed the highest removal during primary treatment:
    a. Primary RR: 0.06
    b. Secondary RR: 0.53

In addition, the fibres and films with large size and low density were efficiently removed by flotation and grease removal, and pellets with high density tended to sink and were removed through gravitational settling. Enhanced removal of PE and PS were done through electrostatic attraction positively charged microplastics and negatively charged activated sludge. Furthermore, the study conducted research mass of microplastics being removed, the analyses shows that WWTPs are highly effective in removing microplastics such as in the influent microplastic masses sees reduction from 1000 mg/L to close to 5mg/L in the effluent section. Overall, less than 3% of influent microplastics were discharged to the environment and a small portion was the microplastic was retained in the sludge as the majority of them are removed through degradation, membrane retention and skimming.

Overall the summary of the literature review is primarily on the concession that the primary treatment removes large and fiber materials through flotation and settling, secondary captures the smaller particles through activated sludge. Advanced filtration systems such as membrane filtration contribute a small fraction at the removal of microplastics.

Nanoplastics removal in WWTPs and DWTPs:

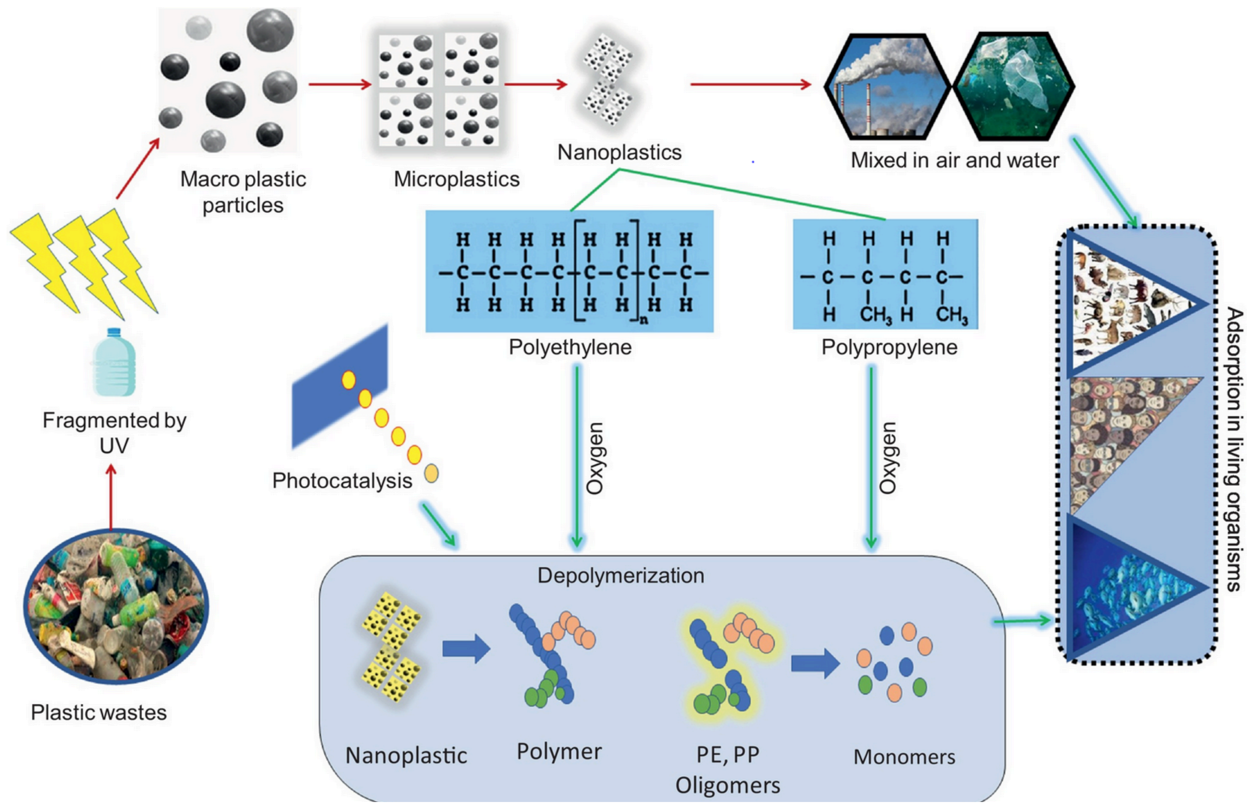


Figure 5. Nanoplastics filtration (Singh et al., 2025)

Nanoplastics are defined as particles less than 1μm, they are becoming more prominent due to their pervasive presence in the environment and increased level of risk to both animal and human health. Their extremely small size leads to better attachment towards organic, heavy metals and microorganisms leading to transportation of them outside to the environment. The WWTPs and DWTPs both act as sources of nanoplastics and then nanoplastic act as a medium to carry out particles to aquatic environments. Furthermore, the sewage treatment plants act as a middle man for the source of plastic pollution. The untreated sewage can transport plastics to rivers and oceans when the treatment process is deemed insufficient. One of the prime examples is that the plastics including NPs undergo shear stress forces during the STP (sewage treatment process) that allow the plastics to fragment and cause surface erosion. This event is not widely known and there has been less research on reporting these fragmentations. NPs because of their smaller size, density and chemical properties, have been characterized inaccurately and led to false positive results. Usually the plastics that are sent to rivers and oceans are microfibres, fragments, beads and films that are less than < 100 µm. (Singh et al., 2025)

Characterization of the Nanoplastics:

In the study of "*Nanoplastics removal during drinking water treatment: Laboratory- and pilot-scale experiments and modeling"* the characterization of the nanoplastics were done through two different batches. The batch 1 (SB1) was used for laboratory experiments, where the nanoparticles had hydrodynamic diameter of 215 nm and polydispersity index (PDI) of 0.04 and zeta potential of 41.1mV. The measurements were then conducted in ultra pure deionized water with the level of pH 6.8. The batch 2 (SB2) was used for pilot plant experiments where the hydrodynamic diameter was 164nm and polydispersity index (PDI) of 0.03 and zeta potential of 4.7mV. The measurements were then conducted in the deionized water of pH level of 6.9. The Zeta potential is the measure of electrical charge on the surface of particles when they are inside the liquid, the higher the zeta potential the higher the particles repel each other. (Pulido-Reyes & Magherini, 2022, #3)

Another characteristic was to calculate the surface area of the NPs through the process of filtration and feed media, the NPs are prepared by collecting water through a filtered membrane of 0.45µm. Then the water is filtered through the sand activated carbon and both the pristine and aged media were studied for the purpose of identifying the grain size of the NPs. The sizes of the media were studied through static light scattering which were as follows:

a. Sand particles (aged and pristine): 450 µm
b. Activated carbon:
    i. Pristine: 1050µm
    ii. Aged: 950µm

Then after the filtration process went through Nitrogen adsorption, which leads to BET method to calculate the surface area of the said particles

a. Pristine sand: 0.58 m^2/g
b. Aged sand: 0.21 m^2/g
c. Pristine AC: 857.37 m^2/g
d. Aged AC: 225.38 m^2/g

Another method to characterize the nanoparticles is through a method called Pd measurement method. This is where the ICP-MS measures the palladium concentration inside the nanoparticles. Solid samples such as sand and activated carbon are treated by acid digestion before analysis and samples like the lake water that contains plastics were directly injected into ICP-MS. The acid digestion procedure for the solid sample that uses a mixture of nitric acid and sulfuric acid. The digested samples were then diluted with water and then Pd concentration was then measured by the ICP-MS. (Pulido-Reyes & Magherini, 2022, #4)

Scaled experiments in the laboratory:

In the ozonation of the nanoparticles, the experiments injected the nanoparticles which were about 1.7 x 10^12 particles/L and then the ice-cooled deionized water was injected with the ozone-oxygen gas mixture. Ozone concentration can be measured by the UV spectrophotometry at 258 nm, and the ozone doses were 0.5 mg/L, 1mg/L, and 5mg/L. Samples were then filtered using the 50 kDa centrifugal ultrafilter, Pd concentration was then filtered using the ICP-MS. This allows the Pd to be quantified and released from the nanoplastics during the ozonation. (Pulido-Reyes & Magherini, 2022, #3)

The filtration process was done in the column with saturated deionized water up in the flow mode to trap the air. Aged media were packed wet to preserve their surface properties and glass fibre wool was placed at the top and bottom of columns to prevent media loss. After saturation, columns were conditioned with lake water. Then the columns were injected with either NaCl tracer solution, or nanoplastic with 1.7mg/L. Nanoplastics transport was evaluated using breakthrough curves and Pd was used as a tracer. BTCs were plotted as C/Co versus pore volumes. The effect of flow rates and columns were tested on as follows:

a. Pristine sand columns
    i. 2 cm
    ii. 11.5 cm
    iii. 20 cm
b. Two flow rates were used
    i. 5 mL/min - simulation slow and filtration
    ii. 21 mL/min - simulates rapid and filtration

In the pilot scale experiments, the area that was to be test was Zurich Water works, two identical down-flow filter columns were used: One for rapid sand filtration, and the other for activated carbon. Sampling was done in 8 different location zones, where 1 port was selected to be at 0.1m above the filter bed, 6 ports at different depths, and 1 port at the filter outlet.

As prior done for injections, NPs were injected to the lake water using the pump, and the concentration of the NPs were 141.6mg/L (0.5 Pd/L). NPs concentration was also adjusted in the filter influent to 1.7mg NPs/L matching very closely to laboratory, then the NPs concentration were injected more to test out the filter aging test and NPs quantification inside the water levels. In addition to injection, the lake water levels were monitored as follows:

a. Lake water temperature: 6 C
b. pH: 7.9
c. Darcy velocity of the water in the column: 1.05 m/h
d. NP injection duration: 6.3h
e. NP-free flushing period: 2.7h
f. Total experiment time: 9 h

During the experimentation, water sampling was collected every 20 min and samples were taken from both side ports and filter outlets. Then solid samples were collected using the metal tube where:

a. AC was 20 g
b. 45g of sand

Solid samples were then acid digested and prepped for Pd tracking. Furthermore testing such as pore volume estimation was conducted every 10 mins at location above the filter bed, within the filter bed, and at filter outlet. Porosity and dispersivity estimation was also conducted.

NPs transportation was then calculated using the transport equation and the main purpose of this equation is that it tells the particle transport in water and particle attachment to filter media.

$$\left\{ \begin{array}{l} \frac{\partial}{\partial t}(\epsilon C) + \sum_i \frac{\partial}{\partial t}(\rho_b S_i) + \frac{\partial}{\partial x}(q_x C) - \frac{\partial^2}{\partial x^2}(\epsilon D C) = 0 \\ \frac{\partial}{\partial t}(\rho_b S_i) = \epsilon k_{a,i} f_{att,i} C \end{array} \right\}$$

Figure 6. Transport Model equation (Pulido-Reyes & Magherini, 2022, #4)

The importance of these parameters is as such:

- **ε (Porosity):** Fraction of pore space in the filter medium available for water flow.
- **ρ_β (Bulk density):** Density of the filtration medium.
- **D (Dispersion coefficient):** Represents the spreading of nanoplastics due to dispersion during transport.
- **q_x (Darcy velocity):** Velocity of water flow through the porous medium.
- **C:** Concentration of nanoplastics suspended in the liquid phase.
- **S:** Concentration of nanoplastics attached to the solid filter medium.
- **k_a (Attachment coefficient):** Controls the rate at which nanoplastics deposit onto the filter media.

For the pilot scale experiment, the nanoplastics explain the short term behavior as well as the long term behavior. In the later stage process (long term), the filter clogging occurred due to particles attaching to the filter surface that led to reduced filter permeability and increased head loss. On the other hand, there was filter saturation (blocking) and only a single layer occurs at the filter, this results in gradual reduction in filter efficiency and complete surface coverage of the filter. (Pulido-Reyes & Magherini, 2022, #5)

The results of the ozone experimentation showed that there is no fragmentation of the particles therefore ozone treatment did not change the NP size under the experiment conditions. Furthermore, the zeta potential decreases as the ozone dosage increased, the values of zeta potential as follows:

- −11.1 mV (control)
- −13.9 mV (0.5 mg O_3/L)
- −15.0 mV (1 mg O_3/L)
- −16.5 mV (5 mg O_3/L)

It was also seen that the low ozone doses used in DWTP also caused decrease in surface change and the following reasons are as follows:

a. Ozone likely caused surface oxidation of the nanoplastics.
b. Oxidation may introduce carboxylic and other oxygen-containing functional groups.
c. The introduced groups make the surface of the particle more negatively charged.

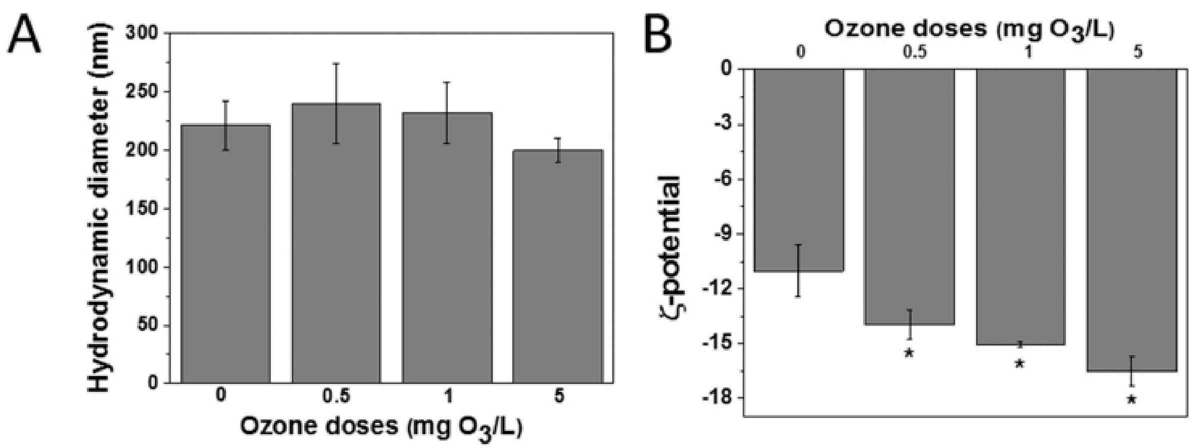


Figure 7. Ozone Results (Pulido-Reyes & Magherini, 2022, #6)

After the ozone was injected, no visible cracks or surface damage were seen, pristine and aged media showed no cracks as well. Therefore, it can be concluded that the ozone up to 5 mg O3/L did not damage the particles. Furthermore, the Pd concentration in the filtered solution was very low up to 0.044 µg/L and means it remained inside NPs during the ozonation. The decrease in the zeta potential was statistically significant but it was below 7 mV. Therefore, such a small change is unlikely to affect the NP transport during the filtration. (Pulido-Reyes & Magherini, 2022, #6)

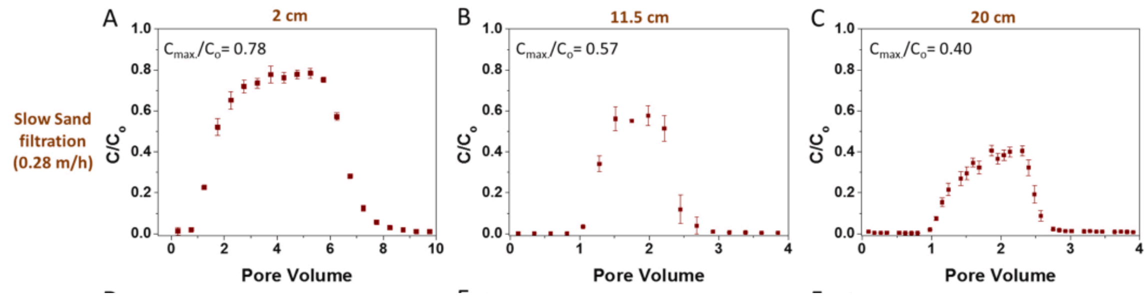
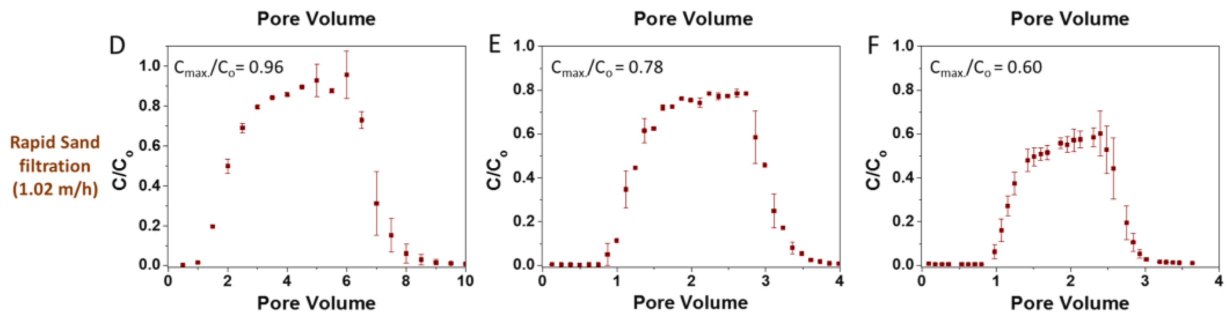


Figure 8. Impact of flow rate (slow sand filtration) (Pulido-Reyes & Magherini, 2022, #6)



F.

Figure 9. Impact of flow rate - rapid sand filtration (Pulido-Reyes & Magherini, 2022, #7)

The figure 9 illustrates that the slow sand filtration rate (0.28m/h) at an increasing filter length reduced breakthrough for nanoplastics. The shortest column (2cm) showed the highest normalized effluent concentration about 0.78 and then increasing the length of the filter to 11.5cm lowered the breakthrough to 0.57. The longest filter length ~20cm showed the greatest removal efficiency but the lowest breakthrough was about 0.48. These results tell us that filter depth enhances plastics retention by increasing contact time and particle media interactions with the filter

For the rapid sand filtration on Figure 9, the nanoplastic breakthrough trend is better than the slow sand filtration. The sand filtration showed an almost complete breakthrough at the effluent concentration of 0.96, while increasing the length to 11.5cm we see a decrease in the breakthrough of 0.78 and then the longest 20cm filter length showed 0.60. Overall, the higher flow rates decreases the nanoplastics retention but increasing the filter length leads to decrease in nanoplastics breakthrough.

Under the sand filtration, the NPs retained in the column were about 22 - 60% depending on their particle interaction. However, as compared to activated carbon, pristine NPs showed higher retention than pristine sand due to its larger surface area. Aged sand also faced higher retention time because biofilms enhanced surface roughness, neutralized surface charge, and created charge heterogeneities. The aged activated carbon instead showed decrease in NPs retention compared to pristine activated carbon, this decrease was mainly due to the reasons of, pore clogging by organic matter, and loss of surface area. Another important aspect to look at is that the lake water increased NP deposition compared to deionized water due to higher ionic strength favored particle attachment by compressing their electrical surface charge. (Pulido-Reyes & Magherini, 2022, #8)

For the pilot scale spiking experiment, sand filters showed more adamant results compared to activated carbon filters showing consistency with the laboratory results. NPs retention were higher in the upper layers with 70% retained at most at 0.1m and 99.5% at the 0.9m depth. Therefore it is highly suggestible that the progressive surface coverage of sand grains can be seen at different sites, however, the activated carbon filter retention increased gradually over time but reached a plateau in the breaking curves.

Types of nanoplastic removal technologies:

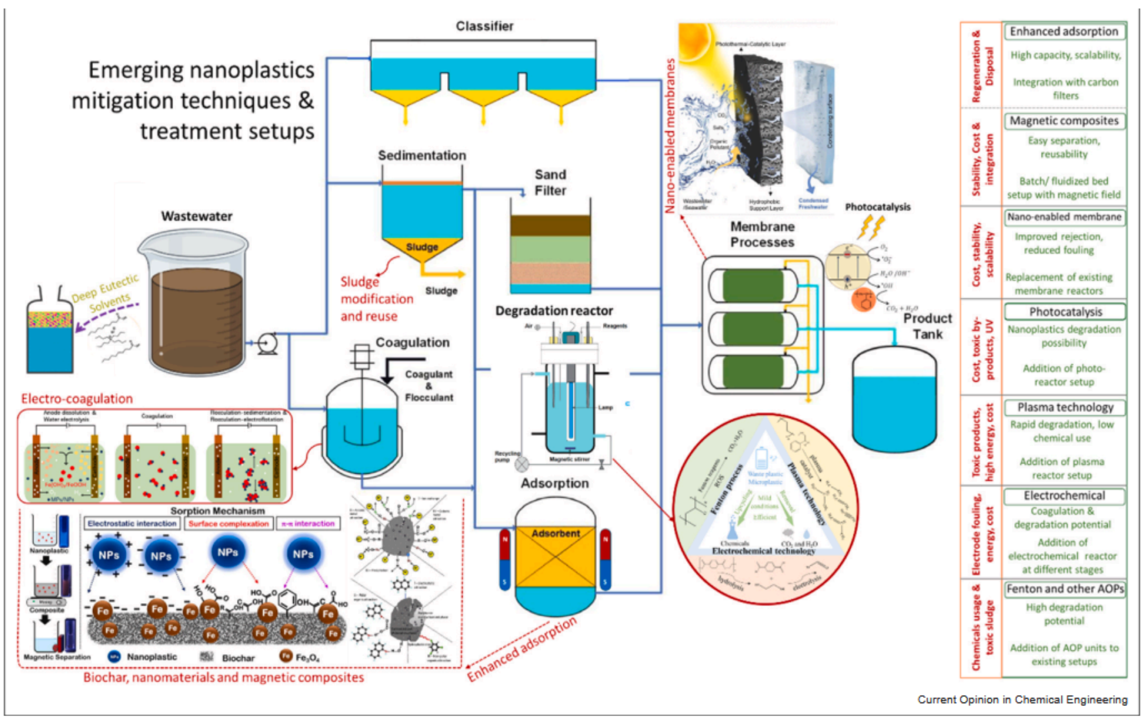


Figure 10. Emerging technologies (Singh et al., 2025)

Nanobased materials:

It is a promising tool that has emerged recently for the removal of nanoplastics, due to their physical and chemical properties. There are various materials such as metallic polymerbased, zeolitic, magnetic ferrite, and metal oxides, which have been explored for mentioned reasons. They work by adsorption mechanisms such as strong surface interactions like electrostatic, hydrophobic interactions that facilitate the NPs attachment. Considering the materials allow large surface area allowing the NPs attachment results in higher removal efficiencies as compared to the conventional media filtration. One of the examples is iron-modified biochar that is able to remove the NPs within 10 mins and provide high ease of magnetic separation. Despite these advantages, the difficulty lies in reusability and potential risk of disposing of these said adsorption materials. Later stages would require future research on environmental impacts of the said adsorption materials. (Singh et al., 2025)

Emulsion based nanoplastic (NP) extraction:

It is also another emerging technology as this method relies on hydrophobic interaction between plastic and the oil phase of an emulsion. Li et al, developed an olive-oil based on an emulsion system that efficiently removed MPs and NPs from different samples, seawater, and toothpaste. After the demulsification, plastic particles were broken and easily separated, therefore, the method worked. One of the issues with this technology is that the integration of the emulsion into the conventional methods requires optimization and painstaking retrofitting due to fouling regeneration affecting the efficiency of the technology. (Singh et al., 2025)

Conclusion:

This literature review highlights the importance of microplastics and nanoplastics removal in the waste water treatment plant as well as the drinking water plant. That said, there is more research that needs to be done on nanoplastics as compared to microplastics due to challenges of size and high mobility. Laboratory and pilot experiments demonstrate that there is certain success in sand and activated carbon media of filtration adjusting the core parameters, flow rate, filter depth, surface aging and media type. Future research should focus on integrating advanced technologies such as detection mechanisms, modelling approaches, and adjusting the environmental risk factors for disposal.

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